

Manual

Monitoring of Process Data PDV-MPD 8.2

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1 Monitoring of Process Data

Overview

Purpose

The PDV-MPD function package includes functions to analyze process data. The values and events collected from the machines may be listed and deviations from default values and limit values that have not been adhered to can be analyzed.

Integration

This function package requires available data in the system (via PDV collection functions).

The licenses and functions for shop floor data collection or batch data collection are required in order to evaluate recorded data based on orders and batches.

Features

- Evaluation of the collection rule values changed by the specification list
- Evaluation of process disturbances for single machines over specified periods
- Collection and evaluation of modifications to target values that have been made, e.g. due to process optimizations. Functions to save changed target values including consideration of past changes
- Process log for individual machines or articles/items over specified periods. Recorded process parameters can be filtered.
- Tabular listing of occurred process events. Events and periods can be selected.
- Tabular listing of durations of occurred events. The durations are calculated by intervals considering the beginning and end of events. Events and periods can be selected. Occurred events can be analyzed based on time and quantity using grouping and sorting functions.

2 Target Value Analysis (in table form)

Summary

Menu	Quality management → Process analysis → Target value analysis
Transaction code	ptva
Function authorization	ptva

The tabular analysis of default values is used to perform process analyses in quality management.

Usage

The tabular view offers the user an overview of the target value changes resulting from manual or automated activities. This designates the limit values and the target value of the characteristics defined in the collection rule.

Requirements

For modifications of the target values from the machine, the machine and the machine interface must transfer these values; and the corresponding assignments to the target values must be made in the logical channels.

Selection parameters

The following selection criteria are available in the application:

Machine

The pool application machine can be used to select the desired machine.

Process parameter

Possibility to enter a process parameter.

Event timestamp (from - until)

By restricting the event timestamp precise time intervals can be selected. In addition, the start and end time of an event may be selected by defining a relative date.

Default type

It is also possible to use a drop-down list to select a default type. These default types may be selected:

- Upper process action limit
- Upper tolerance limit
- Target value
- Lower tolerance limit
- Lower process action limit

Consider long-term data

Enabling this check box allows for long-term data to be considered.

Detail application: target value analysis (in tabular form)

This tabular report shows the changed default values recorded and saved in the database including the following information:

Event timestamp

Point in time when the modified default value was recorded

Machine

Machine at which the default value change was recorded/performed

Process parameter

Technical name of the process parameter for which default values have been changed

Default type

Default type that has been changed

Value

Value of the recorded default value change

3 Violation of Limit Values, tabular

Summary

Menu	Quality management → Process analysis → Tabular violations of limit values
Transaction code	plima
Function authorization	plima

The "tabular violations of limit values" function is used for the process analysis in quality management.

Usage

In this application, the collected limit value violations of the process characteristics are shown. The limits are defined in the collection rule. The table shows the violations and the return to the "OK range" indicating the point in time and the measured value.

Integration

The limits must be defined in the collection rule.

Selection parameters

The following selection criteria are available in the application:

Machine

In the machine pool application, the desired machine can be selected.

Process parameters

In addition, a process parameter can be input.

Event from - to

Restricts the event period by defining the start and end time. In addition, the start and end time may be selected by defining a relative date.

Violation

Restricted by the type of violation using a drop down list

Consider long-term data

Long-term data may also be taken into account by clicking the relevant checkbox.

"Tabular violations of limit values" detail application

This tabular report shows the infringed limit values recorded and saved in the database including the following information:

Machine

Machine where the limit value violation was recorded

Process parameters

Technical name of the characteristic or the entered process parameter

Event timestamp

Time when the limit violation was recorded

Value

Value of the recorded process parameter by which the limit has been exceeded

Violation

Specifies which limit defined within the collection rule has been infringed

4 Process Analysis (based on machines)

Summary

Menu	Quality management → Process analysis → Process analysis (based on machines)
Transaction code	tpdam
Function authorization	tpdam

The process analysis, based on machines, is used for the process analysis in quality management.

Usage

The table view of the detail application offers the user an overview of the existing entries. The display of the information can be sorted using the representation in the grid and is oriented based on the specified selection parameters. The display of the fields can be adjusted for each user with the context menu.

Integration

The function displays collected measured values of process characteristics. These values must have been previously collected in the system.

Selection parameters

The following selection criteria are available in the application:

Machine

A detailed search can be made for the machine using the pool application.

Time domain from - until

Specifies the time interval to be selected.

Process parameters

The process parameters drop down list includes a selection of process parameters.

Tag type + tag ID

Selection of the tag type and tag ID

Consider long-term data

Long-term data is taken into account.

Please note that all data used for these evaluations/reports are kept in the server memory. Consequently, it is not recommendable to select too large periods and data sets. In addition to this, data might no longer be displayed clearly in graphics without additional functions, such as the zoom.

Technical background: every time data is requested in client applications, this data will be stored in the server memory and only then it will be transferred to the client. If you require more memory space, the memory reserved for the Java application needs to be increased accordingly on the server. Please refer to the technical documentation or contact MPDV Support

Detail application: process analysis (based on machines)

This tabular report shows the process characteristics recorded and saved in the database including the following information:

Characteristic

Technical name of the characteristic or the entered process parameter

Designation

Defined characteristic name

Target value

The target value defined for this recorded process parameter at that specific point in time

UTL

The upper tolerance limit defined for this recorded process parameter at that specific point in time

UPAL

The upper process action limit defined for this recorded process parameter at that specific point in time

LTL

The lower tolerance limit defined for this recorded process parameter at that specific point in time

LPAL

The lower process action limit defined for this recorded process parameter at that specific point in time

Unit

Unit of the recorded characteristic

Measured value

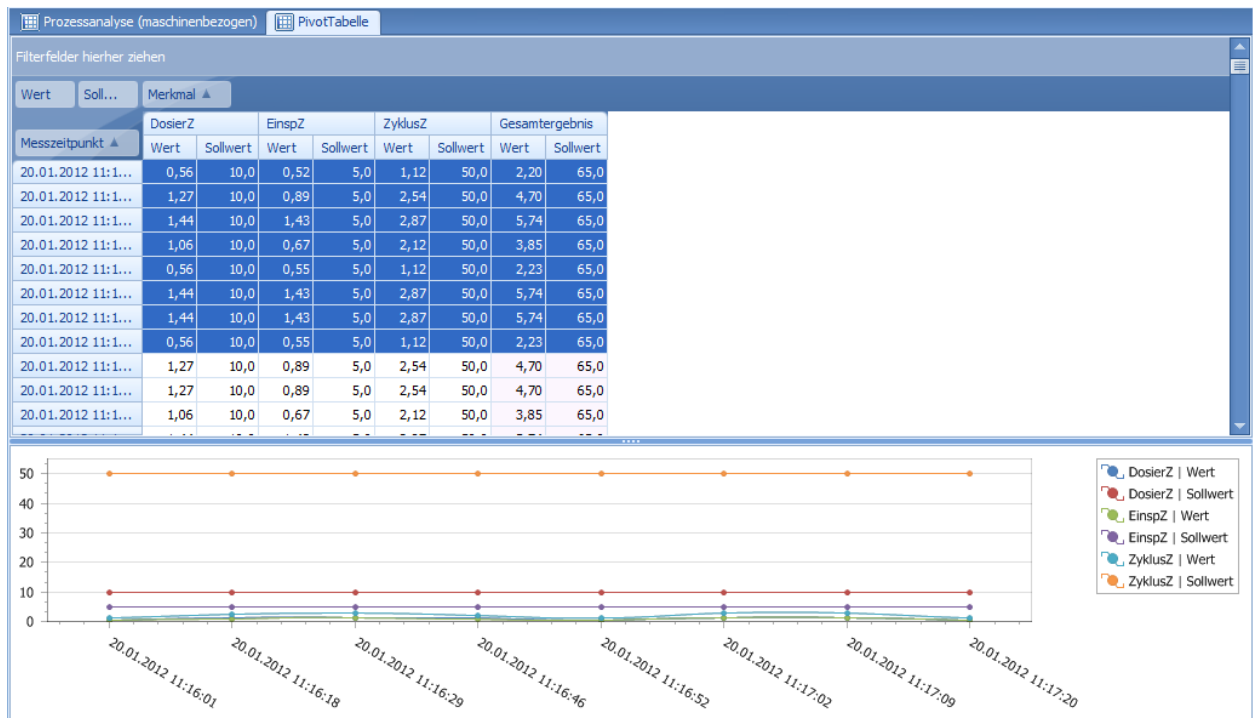
Measured value recorded for the characteristic at the time of measurement

Time of measurement

Time of measurement of the characteristic

Detail application: PivotTable

This report shows the process characteristics recorded and saved in the database in a pivot table including a graphic.



5 Process Analysis (based on orders)

Summary

Menu	Quality management → Process analysis → Process analysis (based on orders)
Transaction code	tpdao
Function authorization	tpdao

The process analysis, based on orders, is used for the process analysis in quality management.

Usage

The table view of the detail application offers the user an overview of the existing entries. The display of the information can be sorted using the representation in the grid and is oriented based on the specified selection parameters. The display of the fields can be adjusted for each user with the context menu.

Integration

The function displays collected measured values of process characteristics. These values must have been previously collected in the system.

Selection parameters

The following selection criteria are available in the application:

MES order number

A detailed search can be made for the order using the pool application.

Article

Article

Batch

Selects a produced batch

Time domain from - until

Specifies the time interval to be selected.

Process parameters

The process parameters drop down list includes a selection of process parameters.

Tag type + tag ID

Selection of the tag type and tag ID

Consider long-term data

Long-term data is taken into account.

Please note that all data used for these evaluations/reports are kept in the server memory. Consequently, it is not recommendable to select too large periods and data sets. In addition, data might no longer be displayed clearly in graphics without additional functions, such as the zoom.

Technical background: every time data is requested in client applications, this data will be stored in the server memory and only then it will be transferred to the client. If you require more memory space, the memory reserved for the Java application needs to be increased accordingly on the server. Please refer to the technical documentation or contact MPDV Support.

Detail application: process analysis (based on orders)

This tabular report shows the process characteristics recorded and saved in the database including the following information:

Characteristic

Technical name of the characteristic or the entered process parameter

Designation

Defined characteristic name

Target value

The target value defined for this recorded process parameter at that specific point in time

UTL

The upper tolerance limit defined for this recorded process parameter at that specific point in time

UPAL

The upper process action limit defined for this recorded process parameter at that specific point in time

LTL

The lower tolerance limit defined for this recorded process parameter at that specific point in time

LPAL

The lower process action limit defined for this recorded process parameter at that specific point in time

Unit

Unit of the recorded characteristic

Measured value

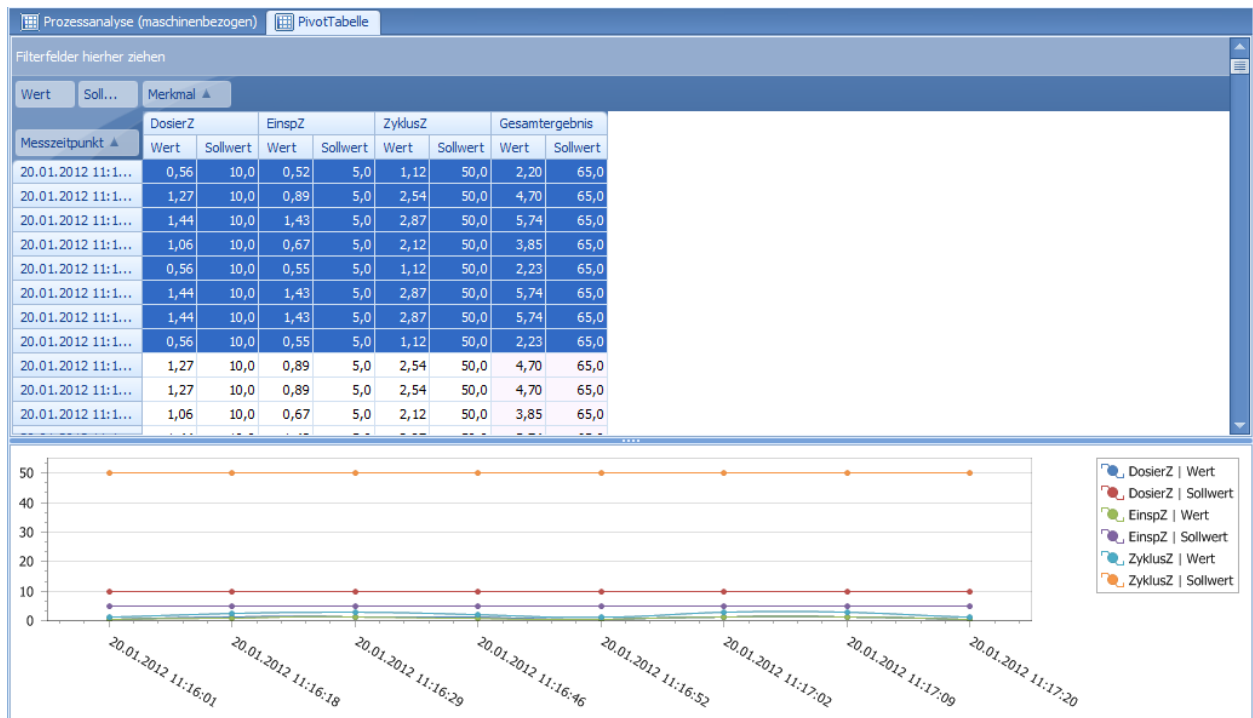
Measured value recorded for the characteristic at the time of measurement

Time of measurement

Time of measurement of the characteristic

Detail application: PivotTable

This report shows the process characteristics recorded and saved in the database in a pivot table including a graphic.



6 Process Analysis (based on batches)

Summary

Menu	Quality management → Process analysis → Process analysis (based on batches)
Transaction code	tpdac
Function authorization	tpdac

The process analysis, based on batches, is used for the process analysis in quality management.

Usage

The table view of the detail application offers the user an overview of the existing entries. The display of the information can be sorted using the representation in the grid and is oriented based on the specified selection parameters.

The display of the fields can be adjusted for each user with the context menu.

Integration

The function displays collected measured values of process characteristics. These values must have been previously collected in the system.

Selection parameters

The following selection criteria are available in the application:

Batch number

A detailed search can be made for the produced batch using the pool application.

Alternative batch number 1

Selects the alternative batch number 1 at the produced batch

Throughput batch number

Throughput batch number

Time domain from - until

Specifies the time interval to be selected.

Process parameters

The process parameters drop down list includes a selection of process parameters.

Tag type + Tag ID

Selection of the tag type and tag ID

Consider long-term data

Long-term data is taken into account.

Please note that all data used for these evaluations/reports are kept in the server memory. Consequently, it is not recommendable to select too large periods and data sets. In addition, data might no longer be displayed clearly in graphics without additional functions, such as the zoom.

Technical background: every time data is requested in client applications, this data will be stored in the server memory and only then it will be transferred to the client. If you require more memory space, the memory reserved for the Java application needs to be increased accordingly on the server.

Detail application: process analysis (based on batches)

This tabular report shows the process characteristics recorded and saved in the database including the following information:

Characteristic

Technical name of the characteristic or the entered process parameter

Designation

Defined characteristic name

Target value

The target value defined for this recorded process parameter at that specific point in time

UTL

The upper tolerance limit defined for this recorded process parameter at that specific point in time

UPAL

The upper process action limit defined for this recorded process parameter at that specific point in time

LTL

The lower tolerance limit defined for this recorded process parameter at that specific point in time

LPAL

The lower process action limit defined for this recorded process parameter at that specific point in time

Unit

Unit of the recorded characteristic

Measured value

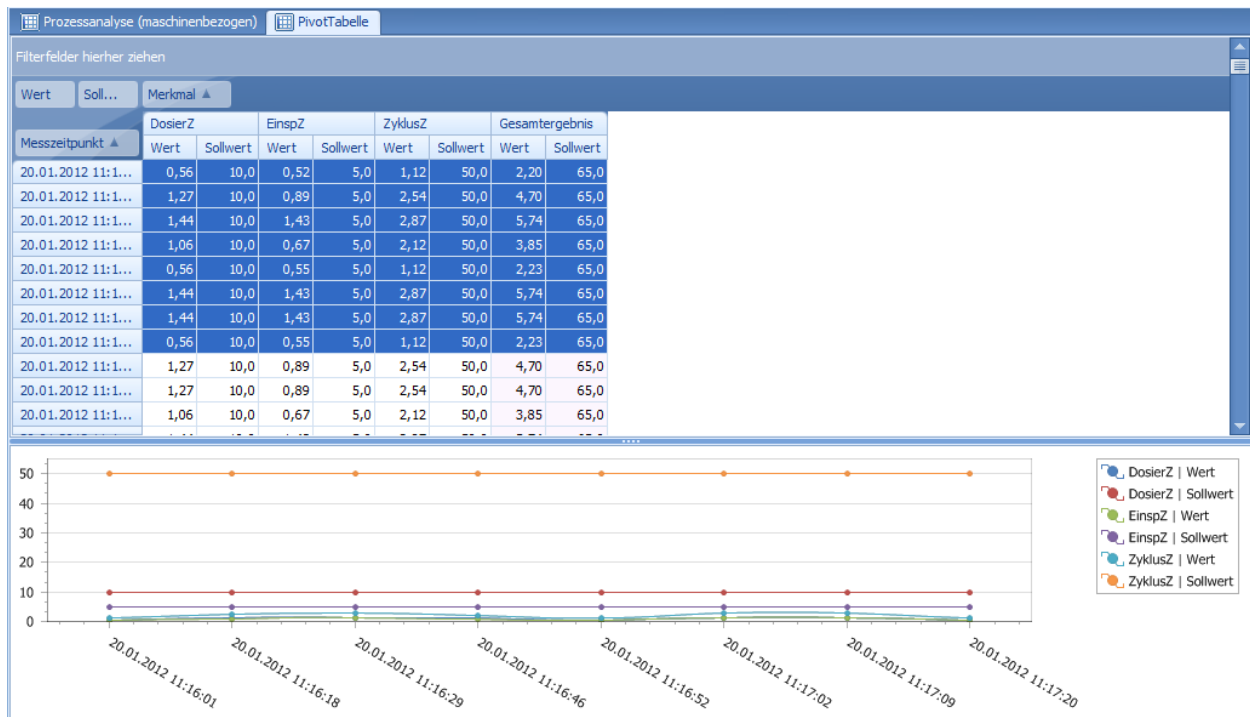
Measured value recorded for the characteristic at the time of measurement

Time of measurement

Time of measurement of the characteristic

Detail application: PivotTable

This report shows the process characteristics recorded and saved in the database in a pivot table including a graphic.



7 Process Events

Summary

Menu	Quality Management → Process Analysis → Process Events
Transaction code	pevea
Function authorization	pevea

This application is used for the event analysis of process events that have been recorded in the process data collection of the machine.

Usage

The application lists the process events in table form. An entry is made for each event regarding whether the event was started (bit value = 1) or ended (bit value = 0).

Integration

The way analog measured values, i.e. process parameters for defined process characteristics are recorded from the machine using PDV, the events are digital process values. These can be unevaluated signals, error signals or message signals.

Prerequisite

The process data collection must be set up for collecting the events.

Selection criteria

The following selection criteria are available in the application:

Machine

The machine to be evaluated.

Event ID

The event to be evaluated or, with wildcards, multiple events.

Event from - to

Limitation of the search for events to this period. The start and end events are selected within this period. In addition, the start and end time of an event may be selected by defining a relative date.

Detail application: process events

This tabular report shows the process events recorded and saved in the database including the following information:

Machine

Machine where the process event was recorded

Short designation

Short name of the machine where the process event was recorded

Designation

Name of the machine where the process event was recorded

Event ID

Technical name of the recorded event

Designation

Event name

Event time

Time of measurement of the process event

Value

Value of the recorded process event

8 Event Durations

Summary

Menu	Quality management → Process analysis → Event durations analysis
Transaction code	pevedrp
Function authorization	pevedrp

This application is used for the event analysis of process events that have been recorded in the process data collection of the machine. These events are evaluated as intervals.

Usage

The application "Event durations" lists the process events in table form. It is assumed here that a start event (bit value = 1) and an end event (bit value = 0) belong together and form an interval. The application shows these intervals and their durations. In addition, by grouping the events, a quantity and total time of occurrence of this event becomes visible.

Integration

The events and thus digital process values are recorded from the machine using the PDV module, similar to analog measured values, i.e. process parameters for defined process characteristics. These can be unevaluated signals, error signals or message signals.

The events displayed in the "process events" application are processed further here.

Requirement

The process data collection must be set up for collecting the events.

Selection parameters

The following selection criteria are available in the application:

Machine

The machine to be evaluated.

Event type

Based on the configuration in the event master data, the type of the events may be filtered here, e.g. "F" or "H".

Period from - until

Limitation of the search for events to this period. The start and end events are selected within this period. In so doing, open end intervals are always calculated up to the end of the selection period. In addition, the start and end time may be selected by defining a relative date.

Consider long-term data

Long-term data may also be taken into account by clicking the relevant checkbox.

Field descriptions**Machine category**

Master data of the machine such as machine, short name and designation

Event category

Master data of the recorded process event

Period category

Start, end and duration of the interval

Grouping:

If grouped by the event designation, the number of events and their total duration are displayed in the total line.